

REEFPRINT/KHANYA: belt mineralogy decision support.

Analogue-data evidence and a staged PGM validation plan

Mintek–SCi Grad Hackathon 2026 · Problem 3: Computer vision for real-time mineralogical characterisation

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October 2026

Abstract

South African concentrators set mill feed, grind and reagents before the laboratory can say what was in the ore. REEFPRINT/KHANYA is a decision-support system that gives the shift team that reading earlier and leaves the decision with a person. Intended product: belt mineral composition; current implementation: recorded-spectrum replay; demonstrated: belt hardness on public analogue data, with the microscope supplementary. In the proposed belt deployment, a short-wave infrared camera would score each 30-second parcel of mill feed for grinding hardness with a conservative upper bound, and Bond's law would turn the bound into a proposed feed rate inside limits set by the metallurgist; so far the hardness model has been tested only by replaying public drill-core data. Belt PGM mineralogy and its $\leq 10\%$ target remain unvalidated. In the shift laboratory, a microscope model maps pyrrhotite, chalcopyrite and pentlandite in polished sections and reports their areas with grain-by-grain evidence; the current method withholds operational advice. An operator would approve, modify or reject every proposed change, the plant's own control system would keep authority, and every step is signed and recorded. All results so far come from public analogue datasets and a plant simulator. On 146 public porphyry-ore composites the hardness model reached an out-of-fold R^2 of 0.479, its 90% upper bound covered 93.2% of samples, and a simulated feed-rate replay estimated a 1.7–1.9% throughput gain over a fixed design-feed policy. The historically evaluated microscope checkpoint reached IoU 0.870, 0.576 and 0.547 for pyrrhotite, chalcopyrite and pentlandite on 12 sections reused during development; the audited application build uses a different checkpoint, which showed large abundance errors on one section. We propose a six-month pilot at one host concentrator with both layers on site, the belt sensor leased and mounted only after a bench test on host ore, costed at R10.69–13.15 million and proposed to be shared between the Technology Innovation Agency (TIA) and the host. Its targets, not yet results, are a phase-abundance error of $\leq 10\%$ on at least 90% of 30 locked microscope specimens, the same target for belt mineral estimates against laboratory mineralogy on locked belt composites, and a hardness bound that holds on the host's belt; if its gates pass, it aims to generate evidence for independent assessment toward technology readiness level (TRL) 5–6 for the microscope (now a provisional laboratory proof of concept) and 6 for the belt system (now 3). We ask first for a bounded pre-contract feasibility stage; the full pilot stays conditional on its result and a host agreement.

1 Introduction

As Team Sonar, working on Mintek's Problem 3, our responsibility is to give a concentrator's shift team a reliable reading of its ore before that ore reaches the mill, and to turn the reading into a plant setting that a person can approve. This report sets out what we built, how we tested it, what it would cost and earn, and, centrally, the six-month pilot that would test it on South African ore (Section 7). It describes a research prototype: the results are on public analogue data and a simulated plant. It is written for metallurgists, mineralogists, process engineers and the people who fund their technology. Technical terms are defined where first used; TRL is the standard technology readiness level, from 1 (basic principles) to 9 (proven in operation) [1].

1.1 Problem statement

The brief states the problem plainly: “without real-time feedback, processing plants cannot adjust to changes in ore quality, leading to inefficient chemical usage and lower mineral yields” [2]. Ore reaches the mill within minutes of leaving the belt, whereas assays and automated mineralogy are batch laboratory work. Mintek's own Minerals Processing and Characterisation Cluster runs QEMSCAN automated mineralogy [3], and a commercial laboratory describes mineralogy as “not an overnight assay service” [4]. Consequently, each shift sets the mill and the flotation circuit for ore whose mineralogy it has not yet seen.

The cost of deciding blind falls most heavily on South Africa, which mined an estimated 120 of the world's 170 tonnes of platinum in 2025 (about 71%) and 70 of 190 tonnes of palladium. It also holds 63,000 of the world's more than 76,000 tonnes of PGM reserves, mostly in the Bushveld Complex [5]. The same source attributes the 9% fall in South African PGM output in 2025 to lower palladium prices, higher deep-level mining costs and disruptions to the electricity supply.

Recovery is where the money is. UG2 ore carries about 30% Cr_2O_3 against about 0.1% for Merensky ore, and the smelter caps the chromite in concentrate. Mintek testwork made UG2 concentrate at 87% recovery with 2.9% Cr_2O_3 , and recoveries above 90% were possible only if the Cr_2O_3 limit was relaxed [6]. Producers report recovery moves of this size to investors: Valterra raised concentrator recoveries by 1.2–1.7 percentage points at Mototolo and Amandelbult in 2025 [7]. Plants do not decide entirely blind: operators, advanced process control (APC) and the mine plan already carry information about the ore. Thus, the decision we aim to improve is narrower: setting mill feed for each parcel of ore before it arrives, with a stated uncertainty, at a cost a producer can recover.

1.2 Proposed solution: two measurement cadences, one operator workflow

REEFPRINT/KHANYA reads the ore at two cadences and passes both to one operator workflow (Section 3):

1. **the belt** (REEFPRINT), every 30 seconds: a short-wave

Table 1: The brief and the judging criteria: evidence and status. Shown = on public analogue data; Partial = part shown; Proposed = future work; Target = defined, not tested.

Requirement	Evidence (section)	Status
≥ 3 mineral phases	Three sulphides on analogue micrographs (4.3)	Shown
Accuracy report	Belt R^2 , bound coverage; microscope per checkpoint (4.1–4.4)	Partial
$\leq 10\%$ error	Target defined; untested on independent specimens (2.6)	Target
Adjust a plant parameter	Feed-rate proposal (designed); historical simulated regrind (2.3)	Partial
Real time	92 ms scoring only; end-to-end latency unverified	Partial
Local validation	Six-month pilot (7)	Proposed
Innovation	Bounded input that abstains, operator review (2)	Shown
Feasibility	Pilot, costs, funding, risks, TRL (7–8)	Proposed
Impact	Value per 1% more metal (6.4)	Scenario
Technical execution	Out-of-fold tests, baselines, failures reported (App. A)	Shown

infrared (SWIR) camera above the mill-feed conveyor feeds a model that predicts each parcel’s grinding hardness with an upper bound, and Bond’s law turns the bound into a mill feed-rate proposal. Today the belt layer estimates hardness; extending it to parcel mineral composition is the intended product and a pilot question (Section 7.2), not a current capability;

2. **the laboratory** (KHANYA), once a shift: a microscope model classifies sulphide minerals in polished sections, which addresses the brief’s three-phase requirement on slower, prepared samples;
3. **the decision**: an operator reviews each proposed change before the ore reaches the mill; the site’s existing control system keeps authority, and every step is signed and recorded.

It is important to note what is not claimed: real-time mineralogy on the belt, performance on South African ore, a measured plant benefit, or identification of chromite and the silicates. Table 1 maps the brief and the judging criteria published in the hackathon FAQ (2025 edition; we have not seen a 2026 rubric) [8] to the evidence, with the status of each.

2 Methods

2.1 Data

No South African belt scans or labelled Bushveld micrographs were available, so every model was trained and tested on public data; none was supplied by Mintek or a producer. HIDSAG (CC0) supplies visible (39-band) and SWIR (75-band) scans of 146 Chilean porphyry copper–molybdenum drill-core composites with laboratory Bond work-index values [9]. A set of 1,112 Bushveld chromitite assay intervals (CC BY 4.0) was used to check PGE routing [10]. LumenStone S2 (research use only; 49 micrographs, 37 for training and 12 publisher test sections reused in development), a Norilsk nickel–copper sulphide ore [11], shares the mineral assemblage of Bushveld ores but not their abundance: sulphides cover 62.8% of its sections by area, whereas Merensky ore contains up to about 3% base-metal sulphides and UG2 less [6]. It supplies labels for the relevant phases, because the base-metal sulphides in UG2 are predominantly pentlandite, chalcopyrite, pyrrhotite and pyrite [6]; its much richer sulphide content and different

preparation can still cause transfer failure, which only host specimens can test.

2.2 The belt layer: hardness from the infrared

Sensor (design). The design uses a SWIR line scanner such as the Specim SX25 (960–2500 nm, 640 pixels across; 162 lines a second assumed, to be confirmed for the configured unit) with a visible–near-infrared (VNIR) companion camera [12]. On a 1.2 m belt at 2 m/s it records a line every 12.3 mm with a 1.88 mm sampling pitch across the belt (not an optical resolution), so it measures averaged parcels of ore, not particles. In the pilot design a *parcel* is 30 s of belt (about 60 m, or 4,860 scan lines averaged into one spectrum); in the replay reported here, a parcel is one HIDSAG composite scan of 80×117 pixels.

Model. Each parcel’s spectrum is reduced to features from the infrared absorption regions of clay, mica, chlorite and carbonate minerals (Fig. 2), and a model predicts the Bond work index (W_i , kWh/t). Results are reported only for samples the model never saw (out of fold), and a result counted only if it beat simple baselines under pass marks committed to the repository before each run. Five targets and several model variants were examined and only the work index survived, so the result is exploratory selection, not a confirmatory test; HIDSAG has no drill-hole identifiers, so grouped independence of the folds cannot be verified.

Bound and gate. The plant setting uses an upper bound that the true hardness should stay below 9 times out of 10 (a one-sided 90% split-conformal bound [13]), not the best guess. A more data-efficient bound [14] was tried first and failed its pass mark, so the exact bound that the plan named as its fallback is used, and the result is provisional. The 90% is a marginal coverage target that assumes new parcels resemble the calibration data; it is not a per-parcel guarantee and does not certify protection against overload. An out-of-distribution (OOD) check flags parcels unlike anything in training, and their advice is withdrawn.

2.3 The decision layer and its cadence

The decision layer turns the hardness bound into a mill setting that a person can approve. Ball mills are usually limited by motor power, so the tonnes a mill can grind depend on how hard the ore is. Bond’s law gives the specific grinding energy [15]

$$W = 10 W_i \left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}} \right) \text{ kWh/t}, \quad (1)$$

where F_{80} and P_{80} are the feed and product sizes (μm) passed by 80% of the material. At a fixed grind target the feed rate the mill can carry is proportional to $1/W_i$, so the proposal is

$$F_{\text{proposed}} = F_{\text{design}} \times \frac{W_{i,\text{design}}}{W_{i,\text{bound}}}. \quad (2)$$

The ratio removes explicit power and size terms only when power, feed size, product target and circuit efficiency are held constant; the pilot must test that approximation for the selected circuit. For example, a design hardness of 16.6 kWh/t and a bound of 16.9 kWh/t give 98.2% of design feed, inside an envelope set by the site’s metallurgist (85–110% in the demonstration). The demonstration falls back to 100% of design feed; at a site, the fallback is the host’s own operating procedure.

In the proposed cadence a person approves changes, not every computation: under the illustrative 30 s acquisition and

90 s scanner-to-mill transit, a parcel scanned from $t = 0$ to 30 s has its leading material at the mill at $t = 90$ s and its last at $t = 120$ s, so a completed estimate leaves about 60 s; preparation, inference, communication, operator response and actuation must fit inside that, with a margin. Advice expires against the arrival of the material it concerns, and a late estimate is kept only as a retrospective record. These are design assumptions, not achieved latency; belt speed, sensor position, actuator response and acquisition-to-action time (p50/p95/p99, with backlog) are measured at the host. A proposal is computed every 30 s (at most 120 an hour), but the operator sees it only when it differs from the running setpoint by more than a deadband set by the metallurgist (for example 2% of design feed), and may approve, modify within the envelope, reject or escalate. With no approval in time, a flagged parcel or stale data, the advice is withdrawn and the site’s own control policy continues; the system never raises feed on its own. Feed may be cut by up to 10% but raised by only 3% per parcel, and the shadow phase measures the real operator-facing rate before anyone acts. Every event is written to a hash-chained record with conventional (Ed25519) and post-quantum (ML-DSA-65 [16]) signatures; this supports traceability, not the correctness of the measurement. In the historical experiments the microscope’s advice reached a simulated regrind setting over OPC UA [17]; the current method abstains, and the feed-rate path is proposed, not built.

2.4 The laboratory layer: minerals under the microscope

The microscope model is a standard image-segmentation network (DeepLabV3 with a ResNet-50 backbone [18, 19]) that labels every pixel of a reflected-light image as resin, pyrrhotite, chalcopyrite, pentlandite or magnetite. In the historical experiments, only the evaluated checkpoint (SHA-256 de7135a9; full hash in the evidence manifest, Appendix A) could send advice to the simulator. A *Quick* scan classifies six 512-pixel fields of a section; a *full* scan covers the whole section.

Connected regions of the mask are split into grains. For this demonstration, chalcopyrite and pentlandite are treated as payload and pyrrhotite as reject; that assignment is a demonstration assumption for this ore, because pyrrhotite’s value depends on PGM deportment and the host’s treatment objectives. In the historical experiments, a rule-based advisor turned a 2D payload-area proxy (grains at least half payload were “free”) into *grind finer*, *continue*, *verify* or *adjust reagent* hypotheses, with a ± 0.335 verify band calibrated on the 12 evaluation sections (Section 4.4). The current application (region-aware contract version 3, measurement observed-target-support-v3) reports target-area categories instead, records retained target support and each exclusion cause, does not apply the historical margin, and always abstains: operational advice remains withheld pending method validation, applicable calibration, release review and host approval.

2.5 Evaluation method

Belt models were scored out of fold against the strongest simple baseline (the fold mean, the fold median and a colour-only model), with pass marks committed to the repository before each run. The microscope was scored on the publisher’s 12 S2 test sections. They were excluded from network training but reused during development, to compare pipelines, select the

checkpoint and calibrate the advisor band, so we report them as *development-influenced evaluation sections*, not an independent test. Accuracy is the intersection over union (IoU: the overlap between predicted and expert-labelled areas, from 0 to 1), pooled over 103.8 million pixels and averaged per section, with bootstrap confidence intervals over sections. Decision-level agreement runs the same advisor on the model’s mask and on the expert’s mask; a disagreement is *unsafe* if the model gives a confident instruction that the expert mask does not support. This measures sensitivity to segmentation error, not whether the rules themselves are metallurgically right.

2.6 Validation targets and independent protocol (proposed)

The $\leq 10\%$ error target must be defined before samples are collected, because absolute and relative errors differ: a reference of 2 wt% predicted as 3 wt% is a 1-point absolute error but a 50% relative error. We propose that the mineralogist and a statistician sign an acceptance contract naming the decision-critical phases, an absolute tolerance below an agreed quantification limit and a $\leq 10\%$ relative target above it, applied per specimen to every decision-critical phase (our proposed definition, Table 6). Image-area abundance is judged separately from decision calibration and from physical liberation. Microscope image fractions are 2D areas, whereas QEMSCAN modal results are usually mass fractions, and a density conversion alone does not make a polished surface represent bulk mass. So the comparison unit, aggregation and density assumptions are fixed in advance, and a same-surface area comparison comes before any mass estimate.

A blind test needs more than hiding answers: a separate custodian holds specimen identities and laboratory results; the frozen model receives only acquisition inputs and permitted metadata; predictions are hashed before the reference is revealed; splits are grouped by specimen and collection event; and every refusal stays in the denominator. We propose 30 development, 10 calibration and at least 30 locked specimens; a failed model faces a new untouched cohort. The protocol is drafted, not signed (Section 7.2).

3 System architecture

Figure 1 illustrates the system, highlighting that both branches share provenance and operator controls but have their own quality checks. The belt and laboratory layers run on different time scales: laboratory results give slower context and audit evidence, not a check on each 30 s parcel. The belt monitor (REEFPRIENT Live) runs the hardness model, the decision view and the signed record. The microscope workbench (KHANYA) is a web application (React) on a Python service (FastAPI and PyTorch) that runs on an ordinary laptop processor. In local mode it accepts connections only from the same machine and makes no external requests, so local image analysis and evidence review can continue offline in the verified local configuration. No language model computes a measured value or a control value. Each result records the input and checkpoint hashes, a versioned measurement recipe, the sampled-field coordinates and the share of the image actually observed, and is refused if any of these goes stale, so every number can be traced to what produced it.

The historical 92 ms median is a Kaggle CPU scoring benchmark on already prepared input, not camera-to-decision latency, which is unverified. Microscope timings for the

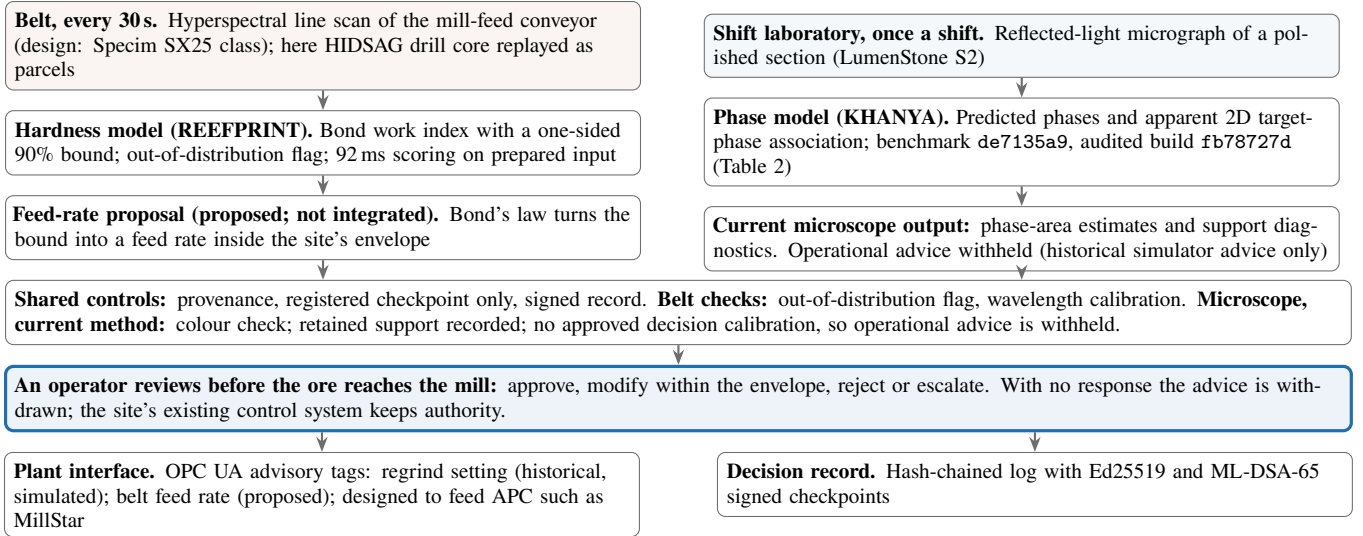


Figure 1: System architecture: two measurement cadences, one operator workflow. The belt and microscope branches produce model estimates at different cadences; the microscope gives predicted phase labels and apparent 2D target-phase association, and independent laboratory measurements provide the reference evidence. Shared provenance and operator controls are supplemented by branch-specific checks. Built: the models, applications and signed record (on replayed public data). Simulated (historical): the plant and regrind interface. Proposed: the belt sensor and feed-rate path. Microscope advice is withheld in the current method.

benchmark checkpoint are log-reported, without retained raw records: 16.7–21.2 s per six-field scan on mains power (5 runs) and 30.8 s on battery; a fresh six-field scan in the audited build (fb78727d) took about 81 s on the audit machine, so the release chosen for the pilot will be re-benchmarked. A static public demonstration runs the belt monitor on precomputed data; a secure server passed 17 of 17 security tests locally and is not deployed. Software verification is in Appendix A.

4 Results

4.1 Belt: predicting grinding hardness

Of the five HIDSAG targets tested (copper and molybdenum recovery, lime consumption, pH and Bond work index), only the work index beat its strongest baseline under every pass mark. Out of fold, the hyperspectral model explains 48% of the variance in work index (R^2 0.479, mean absolute error 1.05 kWh/t), a model given only colour explains 21% (R^2 0.210, 1.29 kWh/t), and the fold mean and median explain nothing (R^2 -0.027 and -0.041). Mean measured W_i is 14.62 kWh/t.

Figure 3 illustrates how the bound behaves, highlighting that it was exceeded on 10 of 146 samples (6.8%). The 95% interval for coverage (88.4–96.6%) spans the 90% target, so the result is compatible with it but does not establish future coverage, particularly on new ore. The price of the bound is width: the bound sits on average 1.64 kWh/t, about 11% of the mean hardness, above the best guess. The OOD check passed 137 parcels, marked 5 borderline and refused 4. For parcel GMET-0004 (Fig. 2) the bound of 16.9 kWh/t against a design hardness of 16.6 kWh/t (the training data’s 90th percentile) gives 98.2% of design feed; the laboratory value was 12.1 kWh/t, so the bound was conservative here.

4.2 Belt: from hardness to a plant decision

Four feed-rate policies were replayed over all 146 composites against a fixed design-feed policy (the 90th-percentile hardness), to show the trade-off between throughput and overloads

(simulated; not PGM ore, not a time series). In the replay a parcel is *overloaded* when its true hardness exceeds the hardness the feed was set for; this is a model quantity, not a plant trip. Setting the feed from the bound, with OOD parcels withdrawn, gave a simulated throughput gain of +1.9% (95% CI +0.9 to +2.9%), and the overload share was 6.8% against 8.9%; the one-sided 95% upper bound on the difference (+1.4 points) is inside the +3-point margin in the evaluation plan. The original replay did not apply the 85–110% envelope; a later analysis with it, reproduced from pinned inputs (Appendix A), gives +1.7% (+0.8 to +2.7%) and the same overload share. Feeding from the best guess instead would add 13.6% but overload 45.2% of parcels, which is why the policy uses a bound. Perfect knowledge would add 14.0%; the bound’s width is where the rest is lost, so a tighter bound from site data is the main route to a larger gain. In the evaluated flotation scenarios grinding faster cost up to 0.25 percentage points of recovery on the steep part of the recovery curve, the largest modelled loss rather than a physical bound, so the site’s curve must be measured.

The result is provisional for three reasons: the exact bound was adopted after the more efficient one failed, HIDSAG has no time order (so transit time, stockpile mixing and the asymmetric ramp limit are not simulated), and the ore is not Bushveld ore. Over 200 random parcel orders, a $\pm 10\%$ ramp limit left overloads unchanged but $\pm 2\%$ added 4.7 points, so feed must be cut fast. A predefined robustness test found the error unchanged for a new capture, half the field of view, added noise and $\pm 15\%$ illumination, but 1.25 times larger, unflagged, under wavelength drift, so a wavelength-calibration check was added.

4.3 Laboratory: phase identification

The microscope addresses the brief’s three-phase requirement on analogue ore. Figure 4 illustrates its accuracy by class, highlighting IoU values of 0.87 for pyrrhotite, 0.58 for chalcopyrite and 0.55 for pentlandite (precision and recall in Appendix A). The pooled mean IoU is 0.5725 (95% CI 0.494–0.624); averaged per section it is 0.4671 (0.412–0.534). The model clearly beats trivial baselines (majority class 0.117;

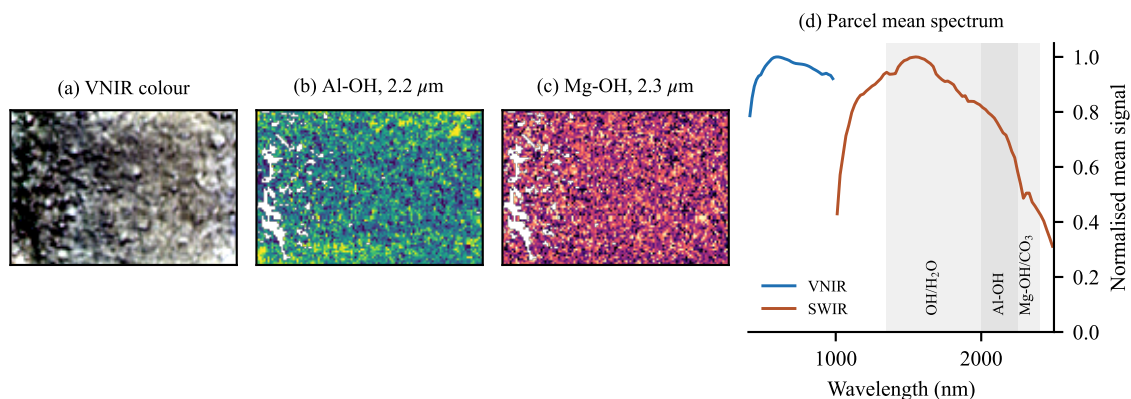


Figure 2: One belt parcel as the hardness model sees it: HIDSAG composite GMET-0004 (Chilean porphyry Cu–Mo), 80×117 pixels. (a) Colour image from the visible bands. (b, c) Relative absorption depth at the Al–OH (white mica, clays) and Mg–OH (chlorite, biotite, carbonate) features; detector stripes removed for display only. (d) Mean spectrum, with the absorption regions the model uses. The model predicts W_i 15.9 kWh/t with a 90% upper bound of 16.9 kWh/t (laboratory value 12.1 kWh/t). The maps show absorption features, not a mineral identification.

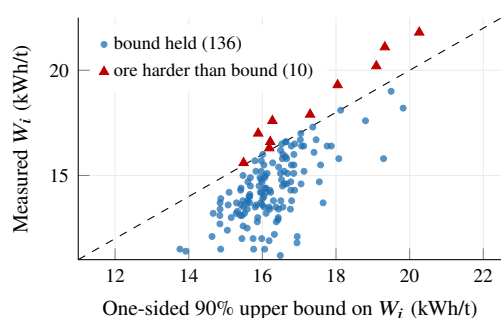


Figure 3: Calibration of the hardness bound, out of fold, 146 composites (no drill-hole grouping available). Points below the dashed line are samples the bound covered: 93.2% (95% CI 88.4–96.6%) against the 90% target. The bound sits on average 1.64 kWh/t (range 1.04–2.14) above the prediction.

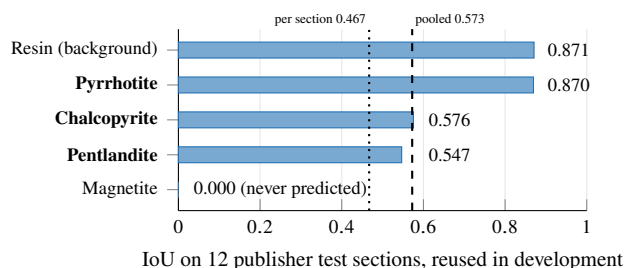


Figure 4: Microscope accuracy by class for the historically evaluated checkpoint de7135a9, full-section inference on development-influenced sections; the pooled mean is over all five classes. The three phases the brief requires are in bold. Per-section means range from 0.340 to 0.728.

colour-only 0.395).

Magnetite, a fourth class the brief does not require, is never predicted. It is only 0.8% of the test pixels: 78% of it is labelled as resin, 17% as pyrrhotite and 5% as pentlandite. The last share is about 0.3% of all pixels predicted as pentlandite, so in this dataset magnetite barely moves the payload measures; a host ore may differ.

Three checkpoints exist and must not share scores (Table 2). The figures above describe de7135a9. The application build audited on 6 October ran fb78727d, which the application itself marks as not approved. On one previously used section (test_11, six fields), the audit compared that model’s output with the expert masks: pixel accuracy was 64.1%, pyrrhotite

Table 2: Model registry. Scores are bound to one checkpoint each. As of 7 October 2026, no checkpoint has been frozen for an independent test.

Checkpoint	Evidence	Status
de7135a9	Mean IoU 0.573 pooled, 12 development-influenced sections	Historical evaluated; research demo
fb78727d	Mean IoU 0.454 (re-run); audit on test_11: pixel accuracy 64.1%, pentlandite 34.8% vs 2.0%	Active in audited build; not approved
42646cfa	Mean IoU 0.632; pentlandite IoU 0.446, precision 0.50	Candidate; quarantined

was predicted at 61.1% of the area against 90.7% in the reference, and pentlandite at 34.8% against 2.0%. This is one development section, not QEMSCAN and not an independent cohort, but it is a concrete counterexample to any blanket $\leq 10\%$ claim for the active model. A retrained candidate, 42646cfa, scores 0.632 overall and detects magnetite, but loses pyrrhotite and pentlandite and has pentlandite precision of only 0.50 (Appendix A); it is quarantined, and the comparative decision safety of the checkpoints is untested. Two further continuations of it on six reused development sections (updating or freezing BatchNorm statistics) lowered pooled foreground IoU from 0.647 to 0.593 and 0.597 and raised pentlandite abundance error from 2.2 to 5.4 and 4.4 points, so both were rejected (continuation bundle of 7 October 2026, recomputed in a separate AI-assisted computational review; not yet in the public repository).

4.4 Laboratory: decisions and grain evidence

In the historical experiments (de7135a9), advice from the model’s masks was compared with advice from the expert masks: the model gave 4 definite recommendations, 6 *verify* and 2 *too few grains* outputs on the 12 sections (action coverage 33%), and none disagreed in the unsafe direction. This small, development-influenced rule-agreement result does not validate the rules or establish plant safety: 0 of 12 bounds the rate over all sections only at 22.1%, and 0 of 4 definite actions only at 52.7% (one-sided 95%).

It is important to note that the ± 0.335 verify band was calibrated on these same sections, so the agreement result is partly

in-sample. A realistic exposure shift (−35 levels on each colour channel) changed the advice on 8 of 12 sections, so illumination control is also a pilot requirement.

The current application implements the relabelling and a border policy: raw edge-connected ore components and overlapping refined components are excluded and counted, and whole-component exclusion can reach far into the image. A development audit of six reused parent images (local evidence, re-checked in a separate AI-assisted computational review) found that even the expert reference masks keep only 0–5.8% of target pixels in Quick mode and 0–70.5% in full mode, so none meets the provisional 80% support policy and all 24 rows abstain; 27,855 target pixels lay on observation edges but 10,263,972 were excluded through connected components (repeated modes, not independent counts). This points to the component-as-particle assumption and specimen preparation, not only segmentation; thresholds are unchanged, and the next study prioritises same-area references and verified preparation.

5 Comparison with literature and limitations

5.1 Comparison with literature

Published work supports the plausibility of the approach, not its validity here. Notole et al. [20] validated hyperspectral imaging and portable XRF against reference methods on Merensky Reef drill core scanned at the Council for Geoscience in Pretoria, with long-wave infrared as well as VNIR/SWIR sensing; that shows the sensing works on Bushveld material, not that our VNIR/SWIR hardness model does, and it is not a public paired conveyor dataset. The three sulphides the microscope classifies are the main base-metal sulphides of UG2 ore [6], and our 50% proxy line coincides with a boundary between middling classes in Mintek’s liberation scheme [3], although that boundary does not by itself define free material.

Four ideas were tested and withdrawn, and they mark where the method stops. The belt spectra did not predict QEMSCAN mineralogy once size fraction and process line were accounted for; a grouped re-analysis (99 records, 36 composites) found only 23–91% of eligible records within 10% relative error, and a process-line baseline beat the spectral model for chalcopyrite and pyrite; an exploratory comparison on the same copper data (B02, Appendix A) narrowed but did not close that gap, so the current belt claim is limited to hardness and belt composition is a pilot question. Belt chemistry routed Bushveld PGE ore worse than the mine plan’s seam label (balanced accuracy 0.79 against 0.82); the belt did not predict recovery, lime or pH; and plant tags never beat the last assay at forecasting concentrate silica.

5.2 Limitations

The following limitations apply, and (a), (b), (e) and (h) block deployment and performance commitments until tested, and justify a separately scoped feasibility investigation: (a) no South African ore has been evaluated, and a model trained on Chilean porphyry core may not transfer, because hardness also depends on silicification, grain size and fracturing; (b) the belt result is a replay of 146 composites without time order or drill-hole grouping, against a fixed design-feed policy rather than the plant’s existing control, and a camera sees the belt

surface, which may not represent the mass entering the mill; (c) belt mineralogy remains exploratory public-analogue development (copper and paired MLA–hyperspectral data), with no qualified moving-PGM-ore validation, and the microscope has not been trained on chromite or the silicates; (d) the 12 microscope sections guided development, and the historical checkpoint cannot be regenerated exactly; (e) the historical advisor’s rules and band are unvalidated for any host ore, and payload-dominance is a 2D proxy, not liberation; (f) only scoring latency is benchmarked, the plant is a simulator, and economic benefit has not been attributed at any site; (g) the active checkpoint differs from the evaluated one and reference-laboratory uncertainty is unquantified; and (h) the latest build withholds microscope advice and refuses the earlier counterexample, but inadequate support even in reference masks leaves specimen preparation, the component-as-particle assumption and calibration unresolved.

6 Business model and financial case

6.1 Value proposition and business model

Our value proposition is simple: the plant gets a reading of its ore before the mill, with a stated bound, and beyond a fixed support fee it pays a success fee only on value that an independent party has measured. No producer, Mintek or TIA has agreed to fund, host or license the work; the roles below are proposals. The buyer is the concentrator or metallurgy manager at a South African PGM producer, who owns recovery, concentrate quality and throughput; named producers are market segments, not customers. Money and value would flow as follows: TIA and the host fund the pilot at cost; Mintek earns from the reference laboratory work and licenses IP through its Office of Technology Transfer (MOTT); the producer receives hardness bounds, phase estimates and the signed record; and an independent auditor verifies the value on which a success fee, capped at 20%, is charged. TIA, the national agency that funds technology towards commercialisation [21], is the proposed main funder (Section 7.5).

The proposed commercial structure has three parts: a capped validation contract; a service fee after acceptance; and an optional variable fee capped at 20% of independently verified net value, with the baseline, measurement period, adjustments, negative-value treatment and audit rights defined in advance. The success fee is zero when verified net value is zero or negative; validation charges and the support fee are separate, so no blanket zero-fee promise is made. Willingness to pay is untested, and laboratory savings are not claimed. The model fits Mintek’s published work [22]: the models need laboratory work, and Mintek’s benefit-sharing model with Hindustan Zinc has generated over R35 million since 2021.

Plants already have cross-belt neutron analysers (bulk elements, with isotope sources that need a licence [23]), slurry analysers (grade after milling), automated mineralogy (phases, in days) and APC such as MillStar and FloatStar. The system replaces none of them; whether it adds ore-side information beyond APC, operator practice and mine-plan blending is what the pilot must show.

6.2 Pilot budget and cash release

Table 4 sets out one scoped planning allowance for the six-month pilot, in two tracks: R6.59 million for the laboratory, team and reference track and R4.10–6.56 million for the belt

track. It replaces the earlier first tranche and separate belt stage; the earlier R8.2–49.2 million installation grid survives only as a superseded sensitivity in Section 6.4. Staff rates are team planning rates; development microscope specimens are referenced by expert annotation, so 44 microscope specimens and the 44 belt composites go to automated mineralogy, because Bond tests cannot validate mineral composition; and every line is replaced by a local quote at G0, with the question narrowed or returned for a new decision if quotes exceed it. The sensor is leased or borrowed with a purchase option where possible, so that failing both bench gates returns it before installation; the belt allowance is a purchase-equivalent ceiling. VAT, duties, major civil work, conveyors or sorters, shutdown costs and closed-loop control are excluded, so the totals are scoped budgets, not worst-case losses. After the pilot, annual support is assumed at US\$30,000–60,000 (R0.49–0.98 million).

The tracks have independent tranches (Table 3), so a belt failure cannot stop the laboratory work. The 44 belt composites split into 20 bench composites, whose Bond and mineralogy references (about R1.05 million) are in L1, and 24 shadow composites, whose references form two ring-fenced tranches even if host-funded: R-H (Bond, about R0.47 million) and R-M (mineralogy, about R0.79 million), each released only for an endpoint that passes its bench gate. A failed endpoint cancels its unneeded orders; references the surviving endpoint needs for baselines and validation remain. If both bench gates fail, B2, R-H, R-M and the belt reserve are never released (avoidable); B1, lease cancellation and the bench references already paid are committed, and the laboratory track continues under its own review. The pre-G0 feasibility stage (P0) is the first funding request: a planning allowance capped at R210,000 over about three months, proposed for TIA or host funding and approved jointly by co-leads and funder: host and stream scoping R40k (co-leads); measurement and decision contract R30k (with statistician); mass and split plan R60k (mineralogist, laboratory); quotes R20k (Hoaeane); eligibility, IP and data terms R25k (Khumalo); reserve R35k. Host access and an eligible funding route are desired outcomes, not promises: if either is not secured within the cap, P0 stops and reports. P0 is outside the pilot total; if it is fully spent, the combined planning envelope is about R10.90–13.36 million before the stated exclusions. The 77 sections include 7 preparation duplicates, so they are not 77 independent specimens. Whether the US\$2,000 anchor covers belt-composite crushing and sectioning awaits vendor scope, and the G0 split plan must show each composite can supply Bond, mineralogy, duplicates and retention. The cost ledger (evidence package) maps each line to tranche, payer, release and cancellation exposure; belt-mineral references sit in the laboratory track for cash control, but their eligibility follows the belt component and is unconfirmed. Committed laboratory orders, staff notice periods and lease cancellation are the costs a stop cannot recover; the payer of each line is agreed at G0.

6.3 From throughput to recovered metal

The belt result is a throughput gain, and revenue comes from recovered metal, so the two must be bridged. For a consistent stream, metal output is proportional to dry tonnes $\times$ head grade $\times$ recovery, so

$$\frac{\Delta M}{M} = \left(1 + \frac{\Delta T}{T}\right) \left(1 + \frac{\Delta G}{G}\right) \left(1 + \frac{\Delta R}{R}\right) - 1, \quad (3)$$

Table 3: Cash release by track (planning allowances; payer agreed at G0). Rows are rounded; the total is summed from unrounded values.

Tranche	Released at	Pays for	Rand
P0	Before G0	Feasibility stage (deliverables in text)	≤ 0.21 M; outside total
L1	G0	Staff and site, months 1–2; kit; inductions; first sections; 20 bench Bond and mineralogy references	1.75 M
L2	Week-8 laboratory review (not G1B)	Remaining staff, site, sections, microscope mineralogy, review	2.72 M
R-H	G1B pass	24 shadow-composite Bond references	0.47 M
R-M	G1M pass	24 shadow-composite mineralogy references	0.79 M
B1	G0	Sensor lease deposit or bench hire; edge computer	0.57–0.90 M
B2	Bench matrix, week 8	Installation, rest of sensor, belt sampling, vendor support	2.71–4.35 M
Reserves	Documented risk event	15% laboratory, 25% belt contingency	1.68–2.17 M
Total			10.69–13.15 M

where T is throughput, G head grade and R recovery. With unchanged grade, a 1.75% throughput gain and a recovery loss of 0.25 percentage points from 87% (a relative change of -0.29%) give 1.46% more metal. Where the plant is limited by ore supply, extra throughput is worth nothing. If all of Zondereinde's output passed the instrumented circuit (an upper bound), 1.46% more metal would be worth R64–123 million a year in margin, and a quarter of that gain R16–31 million; these are simulated gains, not measurements.

6.4 Financial methodology

The following assumptions apply: (a) annual output is that of Northam's Zondereinde mine, 333,050 equivalent refined 4E ounces in F2026 [27]; (b) prices are Valterra's realised basket prices of R32,611 per ounce (2025) and R45,993 (first half of 2026), with cash operating costs of R19,488 and R20,677 per ounce [7, 28], used as proxies for another company's mine, with different baskets, payability and cost structure; (c) for a commercial-life case, installed cost is the belt allowance without the six-month support line (US\$175,000–280,000, R2.87–4.59 million) and running cost is the post-trial support (R0.49–0.98 million a year); (d) the equipment has a 5-year life at a 10% discount rate; and (e) an affected-output fraction f of the mine's output passes the instrumented circuit, taken as 1 unless stated. The annual value of an uplift u in recovered metal is

$$V = u \times f \times Q \times m, \quad (4)$$

where Q is annual output in ounces and m the margin per ounce. For $u = 1\%$ and $m =$ price less cash cost,

$$V = 0.01 \times 333,050 \times (\text{R}13,123 \text{ to } \text{R}25,316), \quad (5)$$

which is equivalent to R43.7–84.3 million a year (R108.6–153.2 million in revenue before costs). Deducting the full average cash cost may be conservative where the mill has spare capacity, but extra mining, handling or bottleneck costs could change that, so a host-approved incremental cash-flow model is needed before investment. The installed cost is spread over

Table 4: One scoped budget for the six-month pilot: team planning allowances, not quotations. US\$ list prices are public anchors from one university facility and one industry guide; rand at R16.4 per US\$ (January–September 2026 average [24]). Every line is replaced by a quote at G0.

Laboratory, team and reference track			Belt track: one belt on site	
Item	Rand	Basis	Item	US\$ (rand)
Two pilot engineers (the authors)	480,000	$2 \times 6 \text{ months} \times \text{R}40,000$	Sensor, optics, references (lease or purchase-equivalent)	100–160k (R1.64–2.62 M)
Software and OT-security engineer	360,000	To recruit; 6 months $\times$ R60,000	Lighting, enclosure, mounting	25–40k (R0.41–0.66 M)
Independent mineralogist	280,000	35 days $\times$ R8,000; annotates development specimens	Edge computer, UPS, network	10–15k (R0.16–0.25 M)
Statistician and data custodian	144,000	18 days $\times$ R8,000	Installation, integration, commissioning	40–65k (R0.66–1.07 M)
Site living, travel, medicals, inductions	318,000	3 people $\times$ 6 months; PPE	Belt sampling crew, references, vendor support	25–40k (R0.41–0.66 M)
Polished-section preparation	154,000	77 sections $\times$ R2,000	25% contingency	50–80k (R0.82–1.31 M)
Automated mineralogy, microscope	1,443,000	44 analyses (10 cal. + 30 locked + 4 repeats) $\times$ US\$2,000 [25]	Belt track total	250–400k (R4.10–6.56 M)
Automated mineralogy, belt composites	1,443,000	20 bench + 24 shadow $\times$ US\$2,000		
Bond ball-mill work index	866,000	20 bench + 24 shadow $\times$ up to US\$1,200 [26]		
Microscope imaging kit	120,000	Allowance		
Independent review, IP and data terms, close-out	120,000	Allowance		
Contingency (15%)	859,000	Released only on a documented risk event		
Laboratory, team, reference total	6,587,000	No overrun without a new decision	Pilot total, both tracks	R10.69–13.15 M

the equipment’s life with the capital recovery factor $CRF = r(1+r)^n / [(1+r)^n - 1] = 0.2638$ ($r = 10\%$, $n = 5$), so the annualised cost $C = CRF \times C_{\text{installed}} + C_{\text{running}}$, which is equivalent to R1.25–2.20 million a year, or R2.50 million with the 25% contingency on the high case. The break-even uplift is then $u_{BE} = C / (f \times Q \times m)$. The more recovered metal needed is 0.015% at low cost and 2026 margin with all output affected; 0.050% at high cost and 2025 margin; 0.063% if the 20% fee is charged before system costs; 0.20% if only a quarter of output is affected or the basket price falls 30%; and 0.52% on the superseded R49.2 million installation grid. If the 20% fee is charged on the margin before system costs, the producer breaks even only when $0.8V = C$. The break-even figures are far smaller than the precision with which a short plant trial can measure uplift, which is why measured value, not break-even, decides G5.

For the pilot itself, R10.69–13.15 million is a scoped planning allowance, not a maximum exposure, because VAT, duties, shutdown and civil costs are excluded and must be assigned to a payer. Tranches (Table 3) leave later money unspent at any stop; the belt range is an allowance pending a lease quote and cancellation schedule, since purchase-equivalent economics do not set lease cash payable. The fee basis must be agreed so that no support cost is charged twice, and measured value needs an agreed metallurgical accounting boundary and attribution method under IP MVP [29].

7 The six-month pilot

As Team Sonar, our pilot problem is this: in six months at one host concentrator, find out whether the microscope meets a $\leq 10\%$ phase-abundance target on independent local specimens, whether parcel-level belt estimates of mineral composition can meet the same target against matched laboratory mineralogy, and whether the hardness bound holds on the host’s own belt, in shadow and then in a short supervised advisory

window. Both layers run on site. A clear negative answer at any gate is a useful result, because it stops spending early.

7.1 Pilot duration and feasibility

The following assumptions make six months realistic: (a) contracting is finished at gate G0, including a quoted measurement-feasibility review of specimen preparation and image–reference correspondence, a leased sensor with a delivery date and a plant stop booked for weeks 8–10, so the clock starts at a readiness event; (b) the sensor is non-contact and mounts above an existing conveyor with read-only data, so installation takes days within that stop; (c) specimens come from the host’s existing shift sampling and belt cuts, so 70 microscope specimens and 44 belt composites (each split for Bond and mineralogy references) over about 16 weeks need only about 5 and 3 a week; (d) reference results return in fortnightly batches within an agreed three weeks; and (e) the software, registry and abundance checks already exist and the build already abstains where support is inadequate, so month 1 goes on measurement feasibility, local data and development-only method selection; calibration is fitted later on its own cohort, and advice stays withheld pending validation, release review and host approval. No host, laboratory or sensor access is arranged yet, so these are G0 prerequisites, and $\leq 10\%$ remains a target. Two limits apply: the sensor is mounted only under the bench outcome matrix (Table 12, week 8), and the four-week advisory window can show the workflow, timing and operator response, but cannot measure a gain of about 2% with useful precision, so independently measured value (G5) needs a longer trial after the pilot. If supply, delivery or the plant stop slips, the pilot is extended; the cohort is never cut to keep the date. The added mineral work must be reflected at G0 in quotes, sample access, laboratory throughput and staff capacity.

Table 5: Proposed mineral decision; “host” marks cells to confirm at P0.

Stream, owner	UG2 or Merensky; host metallurgist (to appoint)
Endpoint	Phases named by host mineralogist (candidates: sulphides; chromite for UG2 [6])
Action, actuator	Investigation alert first; blend or hold only via existing stockpile, blending point or manual procedure (host)
Expiry, check	Expires at material arrival; mineralogy on matched cut
Outcome	Avoided off-specification feed, net of costs (host)

7.2 Pilot objectives and acceptance targets

Table 6 sets out the proposed targets, signed with the host before any evaluation specimen is collected. We use “90% accuracy” in one strict sense for the microscope: at least 90% of locked specimens must meet the error contract for every decision-critical phase. With 30 locked specimens, an observed 27 of 30 shows only that the true pass rate exceeds 76% (one-sided 95%); demonstrating 90% itself needs 29 of 29, or 45 of 46 if one failure is allowed, and these bounds assume independent specimens. So 27 of 30 is a proposed observed acceptance rule, not proof of a 90% pass rate; 46 specimens would add about R0.56 million before contingency. On the belt, 24 composites give exploratory evidence of limited precision (22 of 24 within the contract bounds the pass rate only above 76%), and the exact pass-count rule, handling of every critical phase, refusals and reporting limits are fixed before testing; the sampled cut must match the scanned parcel although the camera sees only the surface, and microscopy acceptance counts are never reused as belt evidence; and because a bound can cover 90% simply by being too wide, its width and refusal rate are judged with its coverage. *Proposed mineral decision (hypothesis to validate)*. Table 5 sets out one candidate decision; P0 chooses a single primary action and confirms the equipment and authority that make it possible. The funded camera cannot hold or redirect ore, an investigation alert is not a conveyor stop, sulphide abundance is not treated as PGM grade, and mineral benefits are kept separate from the hardness throughput simulation.

The pilot then measures whether estimates improve that decision within its deadline, net of operating, verification and error costs; fewer laboratory submissions would be a later, separately validated policy.

7.3 Work plan and daily site routine

Figure 5 illustrates the plan, highlighting that the sensor is tested on host ore before it is mounted, and that the microscope model is frozen before any locked reference is revealed. A typical day on site (proposed; the host sets hours and safety rules) runs: (i) handover, safety talk and sensor checks (references, window, purge air, overnight refusals); (ii) collecting sampler and belt cuts time-stamped to their parcels, logged in the custody register and split for sections, Bond, mineralogy and retention, with the custodian’s script assigning each to development, calibration or locked; (iii) imaging returned sections under controlled light, with locked outputs sealed before any reference is seen; (iv) development work on development data only, comparing shadow predictions with arriving references and, in the advisory window, reviewing each proposal with the operator; and (v) a review with the host metallurgist and signed backups. A reference batch goes out weekly, and a monthly steering meeting checks spend, risks and schedule.

Table 6: Proposed pilot targets. “Agreed” values are signed with the host and statistician before evaluation data are collected.

Target	Pass condition (proposed)	Gate
Phase abundance	Pentlandite, chalcopyrite, pyrrhotite: $\leq 10\%$ relative error above an agreed reporting limit, absolute tolerance below it; ≥ 27 of 30 locked specimens pass, lower bound reported	G1A
Coverage	Refusals and failures stay in the denominator; answered share $\geq$ agreed minimum	G1A
Bench hardness (readiness screen)	20 host composites, split before collection into development and calibration subsets; method chosen on development, gate judged on calibration: error below a metadata baseline; agreed bound width	G1B
Bench minerals (readiness screen)	Same split, with laboratory mineralogy: decision-critical phases beat a metadata baseline on an agreed basis and reporting limit	G1M
Belt shadow	24 belt composites, locked evaluation of a frozen version, no online recalibration. Hardness: predefined count inside the bound, width, refusals. Minerals: $\leq 10\%$ relative error per critical phase above the reporting limit, coverage, grouping by sampling event. End-to-end latency (p50/p95/p99) inside the transit time	G3
Advisory window	Proposals withdrawn when they should be; no setpoint outside the envelope; operator response logged	G4

7.4 Pilot team

Table 7 lists the people the pilot needs. Lethabo Hoaeane and Sibusiso Khumalo jointly lead the pilot with equal decision-making authority, each accountable for their assigned technical workstream (allocation proposed, pending both co-leads’ acceptance). Material changes to scope, budget, milestones, external commitments, model promotion or release need both; neither has a casting vote, and a disagreement pauses the change for specialist or steering advice. The co-leads coordinate evaluation but are not its independent approvers and have no access to locked labels. The software and operational-technology (OT) security engineer must be recruited before G0, and the independent roles are contracted, so that assessment never collapses into self-certification. The host owns plant operation and stop authority; in shadow nothing is written to the plant, and in the advisory window only an operator can change a setpoint.

7.5 Funding structure and the role of TIA

TIA funds technology from proof of concept to pre-commercialisation, with instruments matched to readiness [30]. Its Technology Development Fund lists TRL 4–7 and funds demonstration and pilot plants and techno-economic studies, so it is the route we propose to investigate for the laboratory track; the belt components are currently self-assessed at TRL 3 or below, so their eligibility is unconfirmed and is checked in the pre-G0 work package before any instrument is allocated. Its Seed Fund is listed for TRL 3–8 [30]; one published Seed Fund call, run by the SAMRC for its own staff, sets TRL 5–6 and up to R1.5 million, and asks for a functional prototype, a roadmap, an industry partner and an IP strategy [31]. That shows what an application needs, not our eligibility. The Industry Matching Fund is described for late-stage technologies [30], so it is not assumed for the TRL 3 belt model. Beyond money, TIA links innovators with technology-transfer offices, incubators and investors. The applicable instrument, eligibility, cap and co-funding terms will be confirmed with

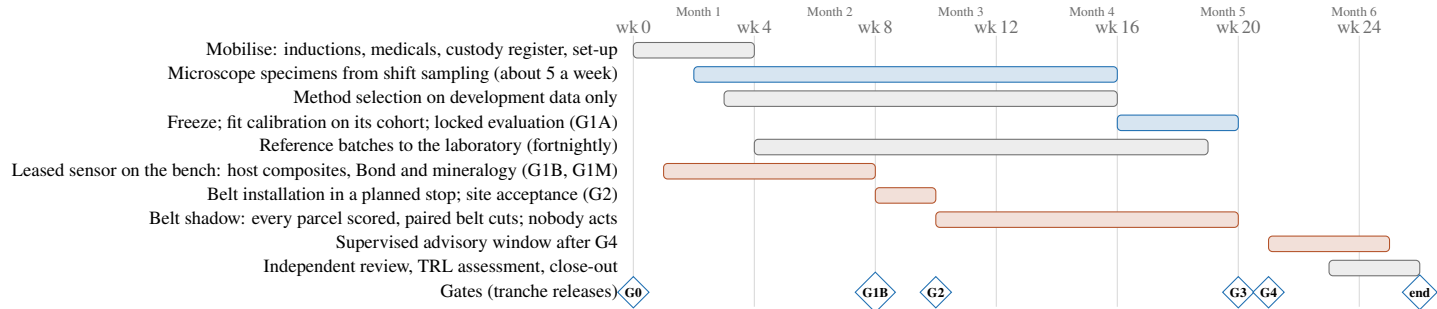


Figure 5: The six-month pilot, in weeks from the readiness event (planning estimate). Blue bars: microscope track; copper bars: belt track. G0 releases L1 and B1; the week-8 laboratory review releases L2; the bench outcome matrix (Table 12) decides B2, R-H and R-M (Table 3). G1A falls in week 20 with G3. If both bench gates fail, the leased sensor is returned before installation and the microscope track continues.

Table 7: The pilot team (functions; one person may cover several, except the independent ones).

Role	Time	During the pilot
Co-lead L. Hoaeane: belt models, evaluation	Full time, on site	Belt models (hardness, minerals), economics; shared planning, budget, host, gates
Co-lead S. Khumalo: microscopy, field, systems	Full time, on site	Microscope, imaging, integration, verification, reporting; shared planning, budget, host, gates
Software and OT-security engineer	To recruit	Sensor integration, offline operation, signed record
Independent mineralogist	35 days	Development annotation, reference review
Statistician and custodian	18 days	Blind split, sealed predictions, evaluation
Vendor engineer	Belt allowance	Installation, calibration, acceptance
Host metallurgist, technicians, operators, safety and OT	In kind	Use case, sampling, envelope, lockout, sign-off

Table 8: Proposed funding structure around TIA (proposals; nothing is agreed).

Stage	TRL	TIA instrument	Cash and other parties
Pilot: laboratory, team, references	4 to 5–6	Technology Development Fund	TIA R5.59 M; host pays Bond tests (R1.00 M with contingency) and gives site access, staff time and laboratory space; Mintek reference work at cost
Pilot: belt	3 to 6	Eligibility unconfirmed; checked before G0	R4.10–6.56 M; team proposes 50:50 with the host (R2.05–3.28 M each), not an agency term
Microscope product	5–6	Seed Fund, terms of the current call	To be confirmed; technology-transfer office
Value trial and roll-out	7–9	Matching or Commercialisation Support	Host-led; investors

TIA and a university technology-transfer office for the current call; Table 8 is a team proposal, with no application made and no amount, split or equipment ownership agreed.

7.6 Pilot risks and mitigations

Table 9 sets out the risks of the pilot period, as the team assesses them, each with a mitigation before it happens and a fallback if it does. Contingency pays only for uncertain costs

inside the approved scope. Three fallbacks run through all of them: the host’s controls keep authority, each stage ends at a gate where spending can stop, and no success fee is due without measured value.

7.7 Governance, compliance and training

These are planned controls, not achieved compliance. The Mine Health and Safety Act requires a risk assessment with the health and safety committee, and training before significant changes to plant [33], which applies to the belt installation; POPIA governs approvers’ names in the record [34]; Mining Charter clause 2.3.1 counts skills and R&D spend [35]; IP from public funding falls under the IPR Act [36], managed through MOTT under the hackathon agreement [8], although the contract settles ownership; and Mintek and TIA spend under the PFMA [37]. Independent checks would be commissioned, subject to agreement and method capability, from a laboratory accredited by SANAS for the selected method’s scope [38], Mintek’s QEMSCAN, a value auditor using a plan adapted from IPMVP [29], an IEC 62443 assessor, an IP attorney through MOTT and a university model reviewer. Each group is trained before its phase, aligned with the site’s skills plan [39]: laboratory staff and metallurgists before G1A work starts, instrument technicians at installation, and control-room operators before G4; competence is shown by observed tasks and sign-off, not attendance.

8 TRL milestones and gates

Table 10 places each component on the standard readiness scale [1] as a provisional self-assessment. Passing a readiness screen or beating a weak baseline does not by itself earn a level; the jumps in Table 10 need an integrated demonstration and assessor review. A level is assessed by an appointed independent reviewer from demonstrated performance against agreed criteria, not awarded automatically at a gate or from elapsed time. The belt system climbs furthest in the pilot, because it moves from public core to the host’s own belt in two steps: host samples on the bench (TRL 4, G1B for hardness, G1M for minerals) and shadow operation on the belt (TRL 6, G3). The advisory window starts operational evidence towards TRL 7; independently measured financial value (G5) is a separate commercial gate, not a TRL criterion.

Table 11 lists what each gate must show, who decides, and what happens if it fails. After the pilot, an extended advisory trial with on/off operating blocks gives the independently measured value for G5, so a commercial decision is about 12 months from the readiness event.

Table 9: Pilot-period risk register: mitigation before the event, and the fallback if it happens.

#	Risk during the pilot	Mitigation (before)	Fallback (if it happens)
P1	Fewer than 5 specimens a week, laboratory turnaround over 3 weeks, or quotes above budget	Sampling plan and turnaround in quotes at G0; second laboratory pre-qualified	Extend collection; narrow or seek a new decision; never shrink the locked cohort
P2	Sensor late, no lease unit, or plant stop missed for installation	Delivery, lease and stop booked at G0; mount tested on the bench	Shift the belt track; shorten the advisory window, not shadow
P3	One or both bench gates fail (G1B, G1M)	Bench tests on host composites before mounting	Apply the bench matrix; if both fail, do not release B2, settle B1 or cancellation; bench references already paid stay with the laboratory track
P4	Belt drift (dust, moisture, wavelength) or scan-sample timing mismatch	Enclosure, purge air, references every cycle; time-stamped parcels	Refuse affected parcels; exclude unmatched pairs and report them
P5	Operator over-trust, unsafe write or cyber incident	Deadband; training before G4; read-only shadow; advisory tags only; IEC 62443 zones [32]	Advice expires or is disconnected; the plant runs on its own controls
P6	The microscope misses the $\leq 10\%$ target, or data leak between sets	One frozen model; independent custodian; sealed predictions	Report the negative result; collect a fresh cohort before any new claim
P7	Engineer not recruited, host withdraws, or data and IP terms unsigned	Recruit and sign before G0; named deputies; LumenStone rights clarified	Hold the next tranche; keep lawful evidence; qualify another host
P8	Belt surface spectra do not represent the parcel's bulk mineral composition	Cuts matched to scanned parcels; grouping by sampling event	Report the negative result; drop the mineral endpoint; hardness continues
P9	Rand 10% weaker on US\$-priced lines; safety incident at the belt	Rand quotes where possible; lockout, escort and permits near moving plant	About R0.64–0.89 M, met from contingency; stop work and follow host procedure

Table 10: Provisional self-assessed TRL now, and the evidence that could support each next level.

Component	Now	End of pilot, if criteria are met	After
Microscope workflow	Provisional laboratory proof of concept; advice unvalidated	5: independent host specimens (G1A); 6: shift routine	7–8
Belt hardness model	3: proof of concept on public core; simulated plant	4: host samples on the bench (G1B); 6: shadow on the host belt (G3)	7
Belt mineral model	Below 3: exploratory copper data; does not beat a metadata baseline	4: bench (G1M); 6: shadow (G3)	7
Decision layer and record	4: built; replayed data; simulated plant	6: shadow; 7 begun in the advisory window (G4)	7–8

Public paired data and planned validation. The product's output is a time-stamped parcel estimate with uncertainty and a valid/invalid measurement status. Public paired MLA–hyperspectral data (204 sections, 49 drillholes, seven locations [40]) have been inspected and used for exploratory site-held-out development experiments. A 20-specimen alignment sensitivity comparison did not materially improve cross-site prediction. Specimen Stonepark bG11A5 has unresolved reference geometry: its historical results are kept with this limitation, it is on hold (exclusion would mean refitting affected folds), and no reconstructed transform is accepted as alignment. Source audits found malformed mask exports, uncertain sensor alignment and label-provenance differences; reproducing export stages (and one full chain) shows computational reproducibility, not physical registration. Infrared fusion awaits registration and calibration evidence, drillhole grouping needs a verified crosswalk, and these sections say nothing about moving PGM ore. Further methods (patch pooling, a laboratory teacher for a belt-only student, unmixing, pretrained features) are added one at a time against the best single model and a metadata baseline. None of this replaces matched PGM belt

Table 11: Gates, decision owners and stop outcomes. Numerical thresholds are signed before evaluation data are collected.

Gate	Release condition	Decides	If it fails
G0	Host letter, quotes, sensor lease and delivery, plant stop booked, data and IP terms, signed contract	Funder, host	Preparation only
L2	Laboratory review (week 8): agreed counts collected and split, reference turnaround within the agreed limit, discrepancies resolved, quotes accepted (values set by agreement)	Co-leads, funder, custodian	Hold L2; fix or narrow
G1B	Hardness transfers to host composites on the bench (week 8)	Reviewer, host	Table 12
G1M	Mineral estimates beat a metadata baseline on bench composites (week 8)	Reviewer, mineralogist	Table 12
G2	Risk assessment, read-only design, site acceptance (week 10)	Host OT, safety	Repair or remove
G1A	Microscope targets met on locked specimens (week 20)	Reviewer, host	Image review only
G3	Belt shadow targets for hardness and minerals: coverage, timing, drift, refusals (week 20)	Metallurgist	Extend or stop
G4	Envelope, change management, training, rollback (week 21)	Change board	No advisory window
G5	Independently measured net value and total cost (after the pilot)	Auditor, host	No success fee

Table 12: Bench outcome matrix (week 8). A minerals-only path cannot claim the hardness-derived 1.7–1.9% throughput simulation.

G1B	G1M	Proposed outcome
Pass	Pass	Both endpoints proceed, subject to G2 and release approval
Pass	Fail	Hardness only; mineral claims and unneeded R-M orders removed
Fail	Pass	Minerals only, after a mineral-driven decision, compatible hardware and a revised value and acceptance plan are approved; otherwise shadow only
Fail	Fail	No installation; leased equipment returned under contract; laboratory track judged separately

and laboratory data.

9 Conclusions and recommendation

The present contribution is a functioning, inspectable evidence workflow and a falsifiable route to local validation. On public analogue data the hardness model improves on simple baselines and the microscope classifies three relevant sulphide phases, but the feed-rate benefit is a simulation, the microscope results are development-influenced, the active application checkpoint has shown large abundance errors on one section, and the $\leq 10\%$ error target is untested. We recommend funding the bounded P0 feasibility stage first. Progression to the proposed six-month programme requires host access, scoped quotations, an agreed measurement and decision contract, and the staged release approvals in Section 7. The $\leq 10\%$ local target remains untested; commercial claims require independent reference agreement, useful coverage and measured net value on the affected stream.

CRedit authorship contribution statement. Lethabo Hoaeane: conceptualisation, belt hyperspectral modelling, belt monitor and decision application, economics, presentation. Sibusiso Khumalo: microscope model and evaluation, advisor and safety gates, OPC UA simulation, verification, technical report.

Declaration of competing interest. The authors declare no competing financial interests or personal relationships.

Acknowledgements. We thank the Mintek mentors for a review session that reshaped the build.

Use of AI tools. AI assistants helped with code and drafting; results link to scripts and evidence in the manifest, planning allowances are labelled, and AI checks do not replace specialist validation.

Data and code availability. Dataset licences are given in Section 2. Public revision:

<https://github.com/Sibusiso-K/KHANYA>; later evidence is local and manifest-bound (Appendix A). A static replay of the belt monitor runs at <https://lethabomh14-reefprint.static.hf.space>.

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A Supporting results

Evidence package:

report/evidence/REEFPRINT_report_evidence_v16.zip and report/EVIDENCE_MANIFEST.md (ledger, replay, hashes, release index). *Microscope* (historical; recall/precision): chalcopyrite 0.716/0.746, pyrrhotite 0.934/0.926, pentlandite 0.742/0.675. *B02* (copper development data, grouped folds): PLS errors 0.28, 0.67, 1.34 points (chalcopyrite, pyrite, quartz), 0.62, 0.98, 3.62 on unseen lines; a metadata lookup did better. *B04*: no extraction recipe replaced the model. *B03* (recorded spectra): the tested local v3.1 build rejects altered intake, recomputes saved estimates and closes the storage defects; software consistency only, not served by the public demo.